

Data presentation, analysis and interpretation

Having designed your experiments and executed them according to your chosen methods (as discussed in Methodology and techniques, this document deals with handling and presenting the data that you have collected. Although I will give you some pointers here, you should consult closely with your supervisor to decide on the best way of using data to address your research aim.

In a T802 dissertation, poor data analysis can lead to unjustified conclusions, which will cause the dissertation to lose marks or even fail. In a practical situation, unjustified conclusions might lead to serious financial, social, environmental or health and safety consequences. As your research progresses and you are faced with having to analyse a lot of data, you need to keep in mind the following points:

- Reaching valid conclusions requires careful analysis of relevant data.
- Conclusions have a context; that is, they apply to a particular situation and are not (normally) universal truths.
- Although you will have considered how you anticipated handling the data in the early stages of your proposal, you will probably need to consider new means of presentation and interpretation when you start your actual analysis.

Although judgement and intuition play their part, it is well worth considering the wide range of techniques available to carry out the organising, presenting and analysing tasks that are needed to produce useful conclusions.

The aims of this document are therefore to:

- explain the role of data collection in primary research
- explain the options for extracting useful information from the raw data
- explain the options for summarising and presenting data in a form that will make it easy to interpret
- explain the process of reaching conclusions
- explain the significance of your conclusions in a wider context than the particular experiment or case study.

I do not go into detail here about the many statistical techniques available, nor indeed into the detail of various techniques for handling qualitative (non-numerical) data. Should you require these techniques (and

the requirements for technological research are numerous and varied), you should consult the books listed in the References.

1 Introduction to data presentation, analysis and interpretation

Part of the process of developing your research proposal goes beyond the activity of collecting the data and deals with what you intend to do with it. There is no point in spending all your research time collecting vast quantities of data, only to find that it does not help you address your research aim. In the early stages of your project, you will probably not consider the precise details of how to process the data, but you do need to be able to show that the data you collect will enable you to achieve the research aim. This implies that you must be able to:

- process the raw data collected
- decide how to present the data, e.g. in tabular or pictorial form
- analyse data using appropriate techniques
- reach conclusions from the analysed data.

This assumes that you will be able to interpret the data in order to come to some conclusions that are both valid and useful. When referring to 'data' here I include various types and formats, including numerical quantities obtained from physical experiments, case-study observations and qualitative data such as the comments made by participants in an interview.

1.1 A note on terminology

The word **data** is often used synonymously with the phrase **raw data** to describe the facts and figures and opinions (of respondents to a survey, for example) collected through primary research.

When the data has been processed, aggregated or arranged in some way such that it gains meaning, it is referred to as **information**. We use information to answer questions (that is, to inform ourselves), and the information may be further processed, combined and compared to produce more information, and so on. Ultimately, the information obtained should enable you to achieve your research objectives, and hence contribute to our **knowledge** of the subject.

The term 'results' is frequently encountered in research reports. Although it is a valid and colloquially useful word, it is used in several different contexts and can sometimes cause confusion. 'Results' can be used to mean:

- data obtained as the outputs of primary research
- results of analysis, implying information
- the successful or unsuccessful outcomes of the research.

For this reason I have not used the term in this document. The word 'findings' is also ambiguous, and for the same reason is avoided here.

Here is an example intended to illustrate the distinctions made.

Data concerning refuse sorting was obtained from households on the quantities of separated glass, paper, cans, etc. collected for recycling, and on people's ability and willingness to sort them. This led to **information** about the extent to which recycling of refuse was being achieved, and how much further education was needed. This in turn led to conclusions about actions that could be taken by the refuse-collecting organisation. Consequent changes in refuse collection would, it was hoped, lead to greater **knowledge** of how recycling rates can be improved.

1.2 The process of reaching conclusions

In the early stages of the research, you need to establish your aim and develop the research objectives that are intended to address the aim. The methodology should produce the data that you will need to achieve the objectives. When you were deciding what data to collect, you should also have been thinking about how you would use it to achieve your research objectives, so as you are collecting the data, you should already have some ideas about how to analyse it.

However, once you have started collecting your data, you may find that it accumulates at an alarming rate, and you are less clear about what to do with it all. So here are the next stages you need to undertake.

1. Process the data so that you have recorded it in a useful format.
2. Select and use a presentation method such as a graph, figure or table in order to arrange the data in a form that will help to derive meaning (that is, provide useful information).
3. Interpret the data to obtain the information that will achieve the objectives.
4. Explain how achieving the objectives addresses the research aim.
5. Derive conclusions about the wider applicability of the research.
6. Suggest where further progress could be made by continuing the research.

Bear in mind that this will be an iterative process; you are likely to go through this cycle two or three times in the course of your research.

1.3 Using information to draw conclusions

When are your conclusions justified by the data? Value judgement may be used in the process of reaching conclusions, such as in deciding which data to select and analyse. However, this must be done as objectively as possible, and result in conclusions that are demonstrably based on the evidence obtained. Conclusions arise from the use of reasoning, carefully accumulated data, some intuition and experience. Data must be interpreted carefully, with due regard to accuracy, relevance and appropriate statistical interpretation. Matters of human judgement are important, but are not in themselves sufficient to make a case without detailed supporting evidence, at least not in the technological context of T802.

So what kinds of data are most widely used in the context of T802 research? Some of the most valuable kinds of data are:

- data obtained by observation
- data obtained from experimentation
- identification of trends

- tests of significance
- tests for correlation
- comments from stakeholders, experts and practitioners.

The final bullet point may seem to lack objectivity, and indeed the comments of one stakeholder or expert may differ from, or explicitly contradict, those of another. However, the collected comments of a representative sample may provide useful evidence for part of the process of forming conclusions. Bear in mind at all times, that if meaningful inferences are to be made from this sort of data, then it must be very carefully collected, and any conclusions drawn must take into account the limitations of the dataset (in particular, see Section 3.3 of Methodology and techniques on sampling).

1.4 Information and conclusions in their context

The quality of information obtained and the validity of conclusions reached are crucially dependent on what you choose to investigate (i.e. your experimental sample, whether that is the performance of a piece of equipment, or responses from participants in a survey). The information and conclusions you derive are bounded by the sample chosen, and must be explained in terms of this context. Consider a practical example:

A prototype of some innovative computer-aided design (CAD) software has been tested for usability with five experienced CAD users as subjects, their backgrounds being in mechanical, civil and electronic (circuit-board design) engineering. Whilst a range of views were expressed, the testing has resulted in a coherent set of conclusions about which aspects were deemed satisfactory and which need improvement. It can be reasonably asserted that these conclusions are valid (that is, true, with high probability) in the context of all engineers experienced in using CAD. The conclusions could not, at this stage, be claimed to be valid in the context of (a) all engineers, (b) engineering trainees, (c) CAD users from non-engineering backgrounds or (d) all computer users, though the validity of the conclusions in these contexts might later be assessed by conducting further tests.

Activity 1

Go to Activity 1 on the use of data and information in your project.

2 Analysis and interpretation of data using basic statistical techniques

The term **statistics** tends to be used in a number of different contexts and with different shades of meaning. The term is usually used to mean either:

- **descriptive statistics**, which is about the use of metrics and visualisations to present summaries of a collection of related data (such as average rainfall by location), or
- **inferential statistics**, which is about using statistical techniques to learn about the properties of a collection of objects from a sample. Such claims are, of course, subject to error, so much statistical

analysis is about determining how confident one can be about the inferences made.

In this section, I briefly consider some of the statistical methods that you might use to present and interpret your data. As ever, the main issue is how to select the appropriate technique for what is required; that is, the appropriate way to view the data in order to answer questions posed or reveal useful information. The danger is that it is easy to be misled if an inappropriate technique is selected. Similarly, overuse of mathematics or statistics can lead readers to believe that your results are better-founded than they actually are. When you use statistical analyses, it is important to consider the following questions:

- Have you considered the nature and completeness of your data selected when deciding how the data should be processed?
- Is the quantity of data sufficient to justify any statistical claims that you are making?
- Is the chosen statistical method appropriate and justifiable in the context being considered?

Bear in mind that there are two possible contexts in which you may be carrying out these analyses: you might be analysing an initial data set to suggest how you will carry on the research, or you might be choosing a means of presenting data in the most informative way for your final dissertation. In these cases, the context will affect your choice of analysis and presentation.

An excellent (and short!) guide to the possible ways of abusing statistics, is Darrell Huff's *How To Lie With Statistics* (1991). This is well worth reading to show how badly choosing or presenting statistics (unintentionally, as well as deliberately) can mislead readers and make unfounded inferences from the available data.

2.1 Basic statistical measures

Arithmetic mean

The very basic concept of the arithmetic mean is fundamental to many a statistical investigation, but be careful in using the term 'average' to refer to the arithmetic mean, as it can also be used for the median value of a set of data (see below). The 'arithmetic mean', or 'mean average' is obtained by adding up all the values of the occurrences and dividing by the number of occurrences. For instance, the mean of the data set 6, 3, 1, 9, 4, 3, 3, 7, 1, 3 is

$$(6 + 3 + 1 + 9 + 4 + 3 + 3 + 7 + 1 + 3) / 10 = 4$$

The mean value may be the principal factor in reaching a conclusion, as in the following examples:

The mean average quantity of recycled paper per household per week was ..., which is commendable.

The mean average temperature recorded for the month was ..., which is unusually cold for March.

The mean average wage for the organisation was ... per hour, which is high for the industry sector.

Yet such conclusions may be of little value because they don't tell the whole story, and may even be dangerously misleading.

- The mean average quantity of recycled paper may have been strongly influenced by a small number of atypical households. Perhaps some houses were also used as home-offices, which produced large amounts of waste and masked the low rates of recycling by other households.
- The supposedly cold month of March might have included a few unusually frosty nights, so that for gardening and agricultural purposes, 'unusually cold' could be a misleading conclusion.
- The wages for the organisation might have included some very low wages, counterbalanced by some very high wages, so that overall the conclusion of 'high for the industry sector' is not convincing.

In such cases, we clearly need more information than just the mean average in order to describe the situation and reach a conclusion that tells a more realistic and useful story. Pictorial representation, for example in the form of a bar chart, may be sufficient. We may also need to look at other calculable factors, in particular the **median**, the **mode** and the **spread** (measured by its **standard deviation**).

Median

Given an ordered set of numerical data, the median is the value that has the same number of values above it as below (for an even number of values, the median is often taken to be the mean of the two middle values, if these are different). For instance, ordering the data set above gives 1, 1, 3, 3, 3, 3, 4, 6, 7, 9; here the two middle values are both 3, so the median is 3.

For a symmetric spread of data, the median and mean will be the same (or at least, very close). In the case of heavily skewed data, the median may be of more interest than the mean, in that it is less likely to be skewed by small numbers of unusually high or low values. Therefore, it is often used to better identify a 'typical' value than the arithmetic mean.

Mode

If one value predominates in a set of data, this is termed the mode. For example, the most frequent value in the data set above is 3, so this is the mode.

The mode may be of use in identifying a strong preference or an overriding numerical value, but is otherwise not frequently encountered. Note that there could be two or more modes, which would be shown clearly on a line graph or bar chart as peaks of equal height, adding more useful information than just the mean and median.

Spread and standard deviation

Given several sets of values, shown in the form of a line graph or bar chart, it could be that they have the same arithmetic mean but are different in the spread of their values. Consider the sets of values A and B:

A: 6, 3, 1, 9, 4, 3, 3, 7, 1, 3

B: 3, 4, 4, 5, 4

Here A and B have the same arithmetic mean, 4, but clearly A has a wider spread of values.

A commonly used measure of the spread of data is the **standard deviation**, which expresses the extent to which data is dispersed away from the arithmetic mean. Arithmetically it is the square root of the **variance**, which is the average of squares of the distances between the arithmetic mean and the data points.

To calculate the standard deviation of the set A above, first calculate the mean average:

$$(6 + 3 + 1 + 9 + 4 + 3 + 3 + 7 + 1 + 3) / 10 = 4$$

Then calculate the variance:

$$\begin{aligned} & [(6 - 4)^2 + (3 - 4)^2 + (1 - 4)^2 + (9 - 4)^2 + (4 - 4)^2 + (3 - 4)^2 + (3 - 4)^2 + (7 - 4)^2 + (1 - 4)^2 + (3 - 4)^2] / 10 \\ & = (4 + 1 + 9 + 25 + 0 + 1 + 1 + 9 + 9 + 1) / 10 = 6 \end{aligned}$$

And finally, the standard deviation is the (positive) square root of the variance:

$$\sqrt{6} \approx 2.5$$

Similarly for set B:

$$\text{mean average of B} = (3 + 4 + 4 + 5 + 4) / 5 = 4$$

$$\text{variance} = [(3 - 4)^2 + (4 - 4)^2 + (4 - 4)^2 + (5 - 4)^2 + (4 - 4)^2] / 5 = 0.4$$

$$\text{standard deviation} = \sqrt{0.4} \approx 0.63$$

So A has a higher standard deviation than B.

The standard deviation is also referred to as the root mean square (RMS) deviation of the data values from the arithmetic mean.

The more the data is located near the mean, the smaller the standard deviation is; the more it is dispersed away from the mean, the larger the standard deviation is. This concept and technique may be of use in the T802 context when considering sets of data that change over time. In such a case the standard deviation as well as the arithmetic mean would be of interest.

2.2 Carrying out statistical tests

The question to ask is not which statistical test should be used, but whether your experiment design was one to which statistical tests can be applied. There are many ways of comparing the results of experiments, and statistical tests can only be properly applied if the experiment design is itself sound. So the most important factor is not doing the statistical tests but looking carefully at the experimental design.

Some introductory information on carrying out statistical tests is available from Rowntree's *Statistics Without Tears* (2000). In general, statistical tests are applied to different sets of data in order to give a measure of whether any differences observed in those sets are a result of a genuine difference, or whether they could be accounted for by the natural random variation in the data. The output of a statistical test is

expressed as a decimal number with a value equal to or less than one (≤ 1). This number represents the probability that the differences between two (or more) data sets are due to chance variations caused by some unknown variable. A probability of 0.1 indicates that the likelihood that a set of results occurred by chance is 10%. The lower the probability score from a given statistical test, the greater the significance that can be attached to the results. Two levels of significance are widely used. A probability of 0.05 (5%) is generally regarded as significant, and a probability of 0.01 (1%) as highly significant.

A wide range of statistical tests are available using the data analysis tools provided with most spreadsheet software, such as Microsoft Excel or the (freely available) OpenOffice Calc.

2.3 Correlation and causality

Correlation describes the degree of closeness of two (or more) measures and the direction in which changes in one measure are reflected by changes in the other. When large values in one measure, e.g. rate of pay, co-occur with large values in the other measure, e.g. some quantitative measure of job satisfaction, this is called a **positive correlation**. If larger values in one measure, e.g. frequency of rest periods, co-occur with smaller values of another measure, e.g. frequency of making errors, this is said to be a **negative correlation**.

Let us look at some simple examples of the correlation between two variables from an imaginary experiment. Suppose we were to measure the length in centimetres and weight in grams of a bunch of carrots. We plot the length and weight of the carrots on a scatter diagram in which one axis is length and the other is weight. Suppose that we discover in doing this plot that the dots form an absolutely straight line so that for each increase in 1 cm of length there is a corresponding increase of 8 g in weight. The plot would look like that shown in Figure 1 and describes a **perfect positive correlation**. A **perfect negative correlation** would result in a perfectly straight line going from a high value on the y-axis to a low value on the y-axis of a scatter diagram.

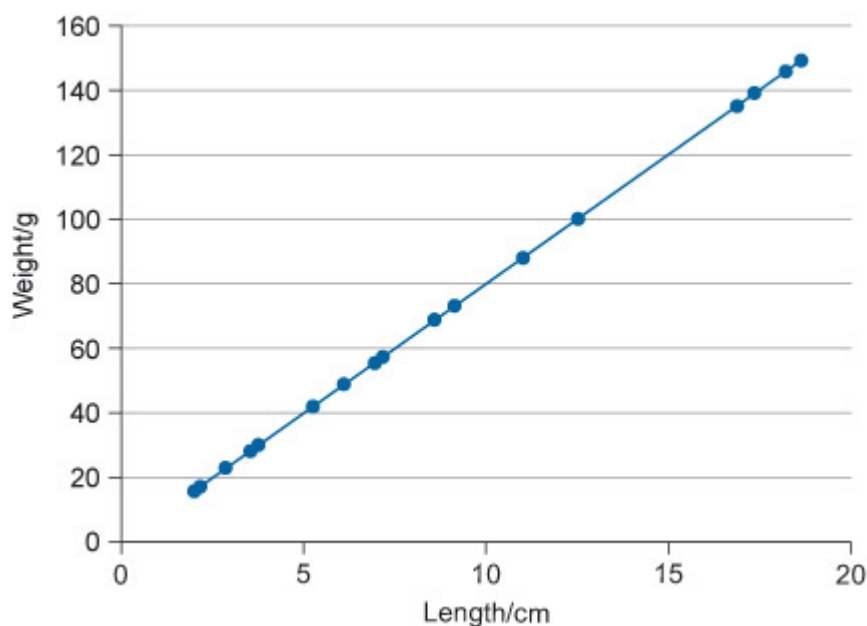


Figure 1 Height and weight correlation in an imaginary sample of carrots

Show description ▼

The scales chosen on graphs of this kind can exaggerate or minimise the degree of change graphed, so they should be chosen carefully to avoid misleading the reader.

Perfect correlations are extremely rare in experimental data, so our plot of the carrots is likely to look (a bit) more like that in Figure 2. If some of the dots fall a little to the sides of this straight line, then there is still a positive correlation, but the correlation isn't perfect.

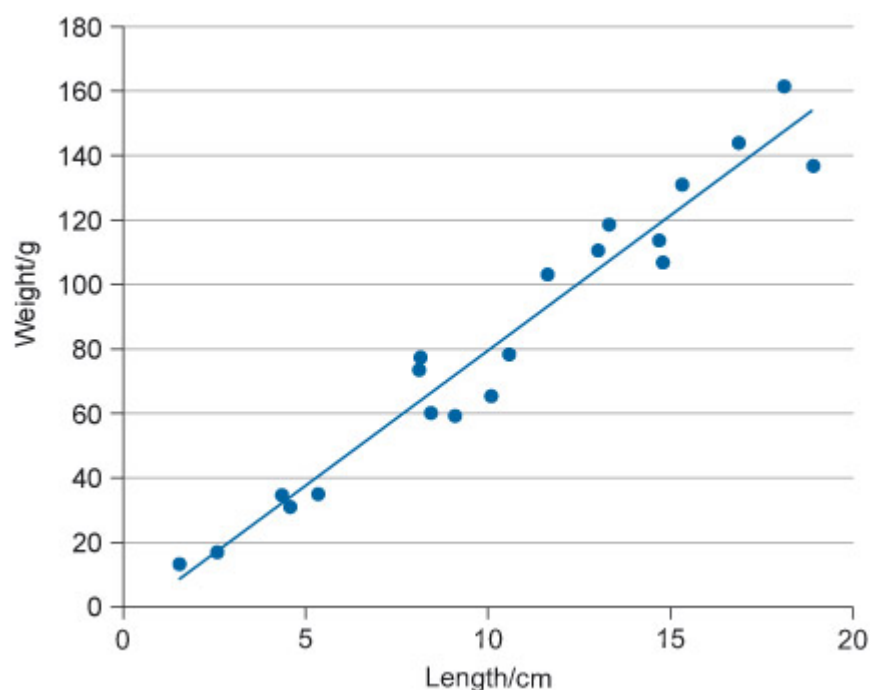


Figure 2 Height and weight correlation in an imaginary sample of carrots

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No correlation exists if points plotted do not form a weak or strong association along a line. Figure 3 shows an example where no correlation exists.

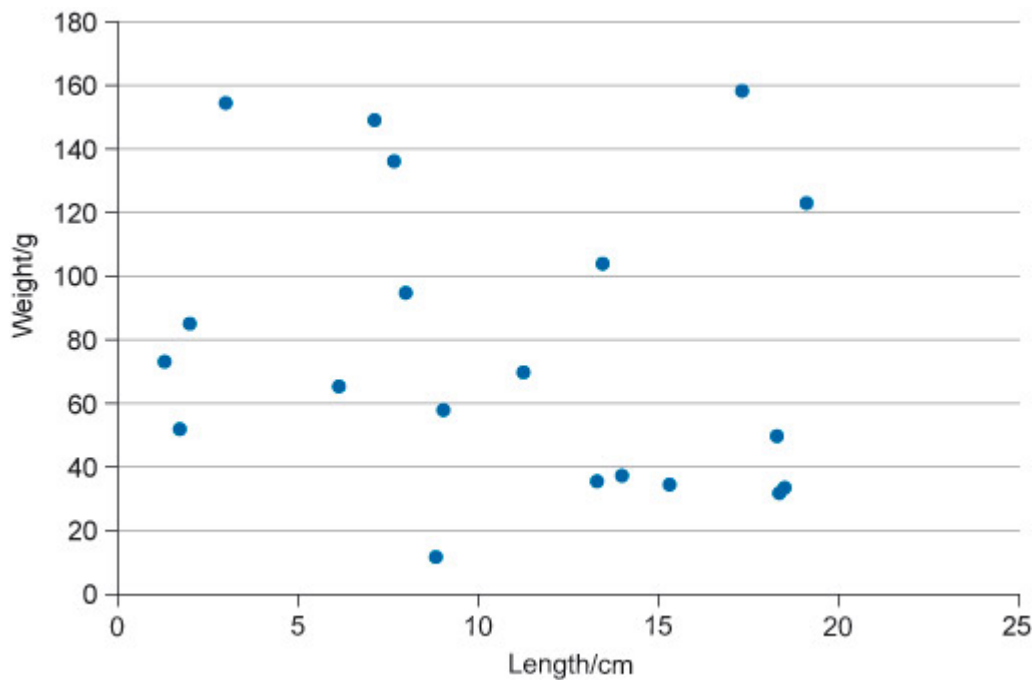


Figure 3 Uncorrelated points

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For further discussions on correlation, including how to interpret measures of correlation, you should consult a text such as Rowntree (2000) or Rees (2000).

It is essential to note that correlation does not imply a causation. For example, while there may be a correlation between higher rates of pay and job satisfaction, this is not to say that higher rates of pay *cause* job satisfaction. Similarly, suppose that in a manufacturing context a strong correlation has been identified between late arrival of raw materials and outsourced components (X), and late delivery of products to customers (Y). It might appear self-evident that X causes Y because of the strong correlation. Indeed it might be true, but it could be that X has a partial or even minimal effect on Y. Other factors, such as a processing bottleneck at a particular machine, poor scheduling of skilled operators, faulty output requiring re-working or many others, might be more significant. To reach conclusions about causes, a researcher would need to look at the detail of product manufacture that resulted in specific late deliveries. The causes of the various delays that contributed to the overall delay could then be identified and prioritised, resulting in a more coherent view of the cause of Y.

2.4 The Pareto principle

A frequently quoted concept is the 80:20 'rule', often referred as the Pareto principle after its originator, in which 80% of something is related to 20% of something else (although you should treat it as an illustrative rule of thumb, rather than a reliable rule about phenomena in the world). A typical example is encountered in the statement '80% of our turnover comes from 20% of our customers' – typical, at least in approximation, in much of the manufacturing industry. However, this is a rule of thumb or heuristic, and by no means a universal truth. For example, in the postal industry, it is questionable whether 20% of customers post 80% of letters. The Pareto principle is commonly encountered, but can be misleading, and at best should only be used as an approximation.

It is clearly unrealistic to assume that the 80:20 distribution applies to all situations. However, it can be useful to investigate such distributions as a starting point in analysis, such as investigating fault/cause situations. We can therefore use Pareto analysis, which is not the same as assuming the Pareto principle, but is based on the search for the most significant factors, those that could be dominant. In a situation where over 50 categories of faults are recorded, but the most frequently occurring fault situations appear to be caused by only a small proportion of the fault categories, then Pareto analysis may be a useful shortcut to establishing where to apply fault correction most effectively.

One approach would be to analyse the fault occurrences and establish the most frequent causes (for example, see the discussion by Schulmeyer and McCabe (1998)). There could be around 10 causes (20% of 50) causing 80% of the occurrences, but there is no natural law that insists on it. Having examined the overall spread of faults it may be useful to examine in detail the top 3, 7, 15 (or whatever) causes. The decision would come from the practicalities and context of the actual situation, not from adherence to an 80:20 rule. Pareto analysis refers here to the identification of causes in descending order of frequency, but could be similarly applied in other distributions of categories.

In summary, Pareto analysis is a useful technique in statistical analysis, but it should not be used with the assumption of an 80:20 distribution.

2.5 Summary of statistical methods

What can we conclude from this discussion? First, bear in mind that this is not intended to be a detailed treatise on statistical methods; statistics is a difficult area, and the resources given in the References will provide you only with an introduction. Here, I want to emphasise the following points.

- Be careful about how you use statistical methods, and in any case only use them if dealing with sufficient quantities of data. Read about your chosen method before using it and be sure you are using it in an appropriate context. If you do not have a background in statistics, you are advised to consult with your supervisor on the most appropriate methods to use.
- Basic calculations of properties such as the arithmetic mean are useful in themselves, but rarely tell the whole story. Presenting the complete range of data in suitable graphical form, and calculating other factors such as standard deviation, can lead to more convincing and reliable conclusions.

Activity 2

Go to Activity 2 on statistical interpretation of quantitative data.

3 Graphical representation of quantitative data

The key to successfully interpreting data is to *select the most appropriate way to present the data*. If you choose well, you will be able to see patterns or trends in your data that may not have been visible from the raw data alone. However, it is easy to choose visualisations of the data which either do not convey a genuine pattern, or which mislead you into seeing a pattern that is not really there.

It is important to bear in mind that you are likely to use representations in two different ways during your project.

1. You will find that using some initial plots is invaluable to help you get a feel for the 'shape' of the data, and to spot initial trends. This use of representations is vital to start pointing to the ways you should go about your final analyses.
2. When writing up your final dissertation, you will use a careful selection of representations that best convey the key points that you want to make.

Point 2 above is *not* intended to suggest that you should artificially select a subset of data or a representation which is most sympathetic to your preferred conclusion. Rather, that your choice of representation needs to be appropriate for the kind of data that you wish to discuss. For example:

- A bar chart is useful for showing the absolute and relative values of discrete variables.
- A time-related graph or a histogram is useful for showing trends.
- A pie chart will readily show up proportions and quantitative categorisations.
- A star or radar chart is a useful pictorial means of comparing several different values of a number of discrete variables.

The figures in the following sections demonstrate the main methods that can be used for graphical representation of data. They can readily be reproduced using the graph drawing options in software such as Microsoft Excel or the OpenOffice suite. They are by no means a comprehensive set, and there are many variants not shown.

3.1 Line charts and scatter diagrams

A line chart can often be the most useful way to show the correlation between two variables. The most common type is the straight line chart (Figure 4), which plots a dependent variable (the vertical axis) against an independent variable (the horizontal axis). Samples are usually taken at regular intervals, but if a more accurate graph is required then a continuous chart may be used (Figure 5).

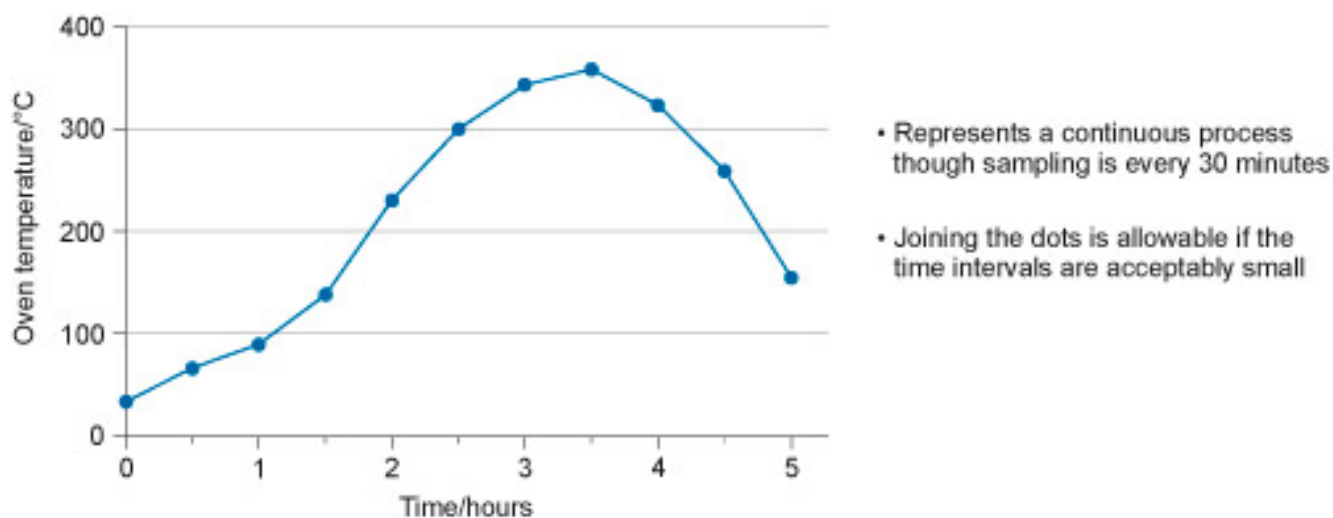


Figure 4 Example of a straight line chart: oven temperature vs time

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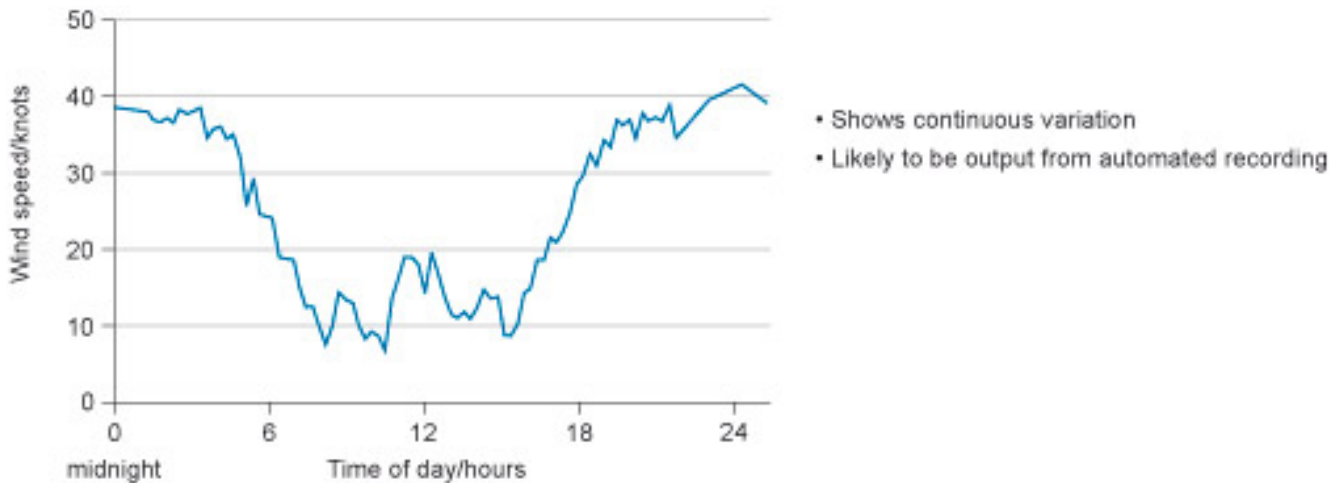


Figure 5 Example of a continuous chart: wind speed variation over 24 hours

Maximise

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A scatter diagram (Figure 6) is also a good way of showing the correlation between two variables. In this case the variables may be dependent or independent; the plot just shows the association (or lack of association) between the two.

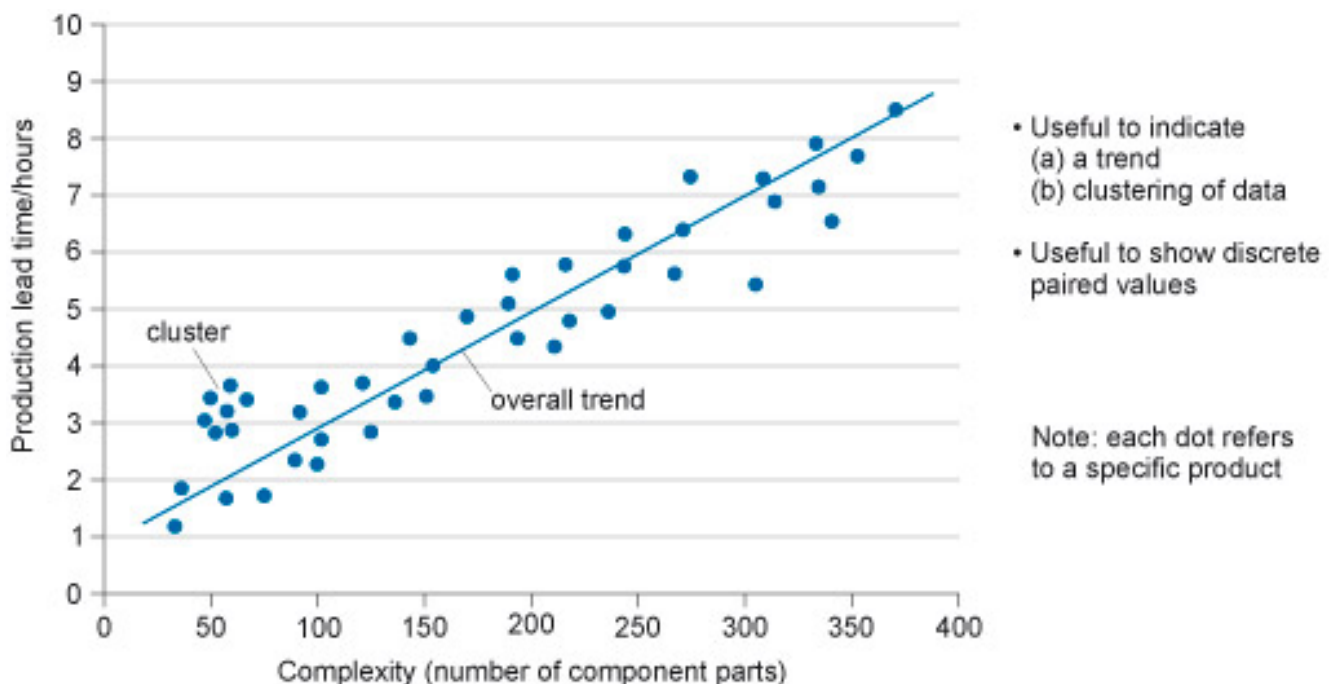


Figure 6 Example of a scatter diagram: plots of lead time vs complexity

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3.2 Bar charts and histograms

It is easy to confuse bar charts and histograms. In general, bar charts are more often used when one of the variables being considered is discrete – for instance, different companies or different countries.

Histograms, on the other hand, are frequently used when the independent variable is continuous; here the variable on the horizontal axis is divided into intervals (which may or may not be of equal size) and the area of the bar represents the information about all the occurrences in the interval.

The main difference between the two types of chart is that in a bar chart it is the height of the bar that denotes the value, whereas in a histogram it is the area. This distinction is not always obvious, since if the intervals in a histogram are of equal size then essentially it is the height that matters. However, it is important to remember the difference when you come across a histogram that has 'bars' of different widths.

Bar charts

A bar chart should be used for showing discrete data, although it can also be useful for revealing trends over time. However, the bar chart does not attempt to present a continuous picture of how the dependent variable changes; instead, it compares values for discrete time periods such as weeks, months or years.

Figure 7 shows effluent discharge rates for each of seven weeks. The implication is that this represents seven different measurements (at one per week). To illustrate the total continuous discharge over a seven week period, a histogram might be more appropriate.

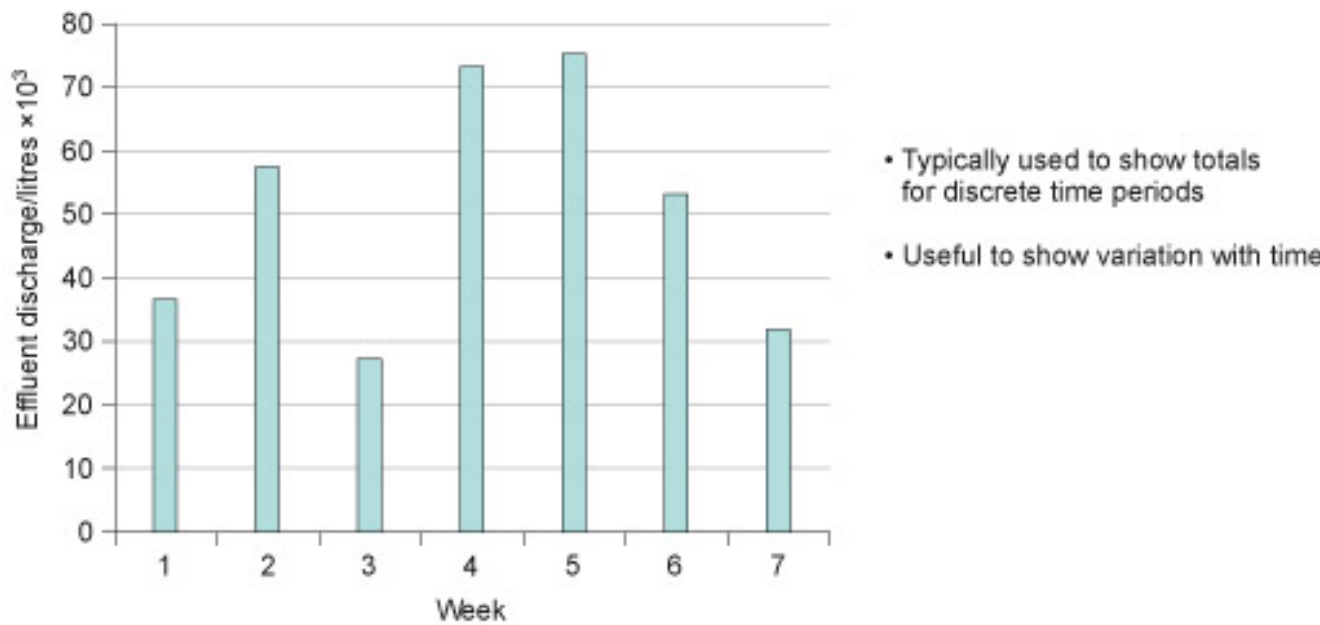


Figure 7 Example of a bar chart: weekly effluent discharge

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A typical spreadsheet will allow you to make many variants of the basic charts, some of which are more useful than others. For example, Figure 8 shows effluent discharge over a period of 5 weeks, compared across the years 2009-2011.

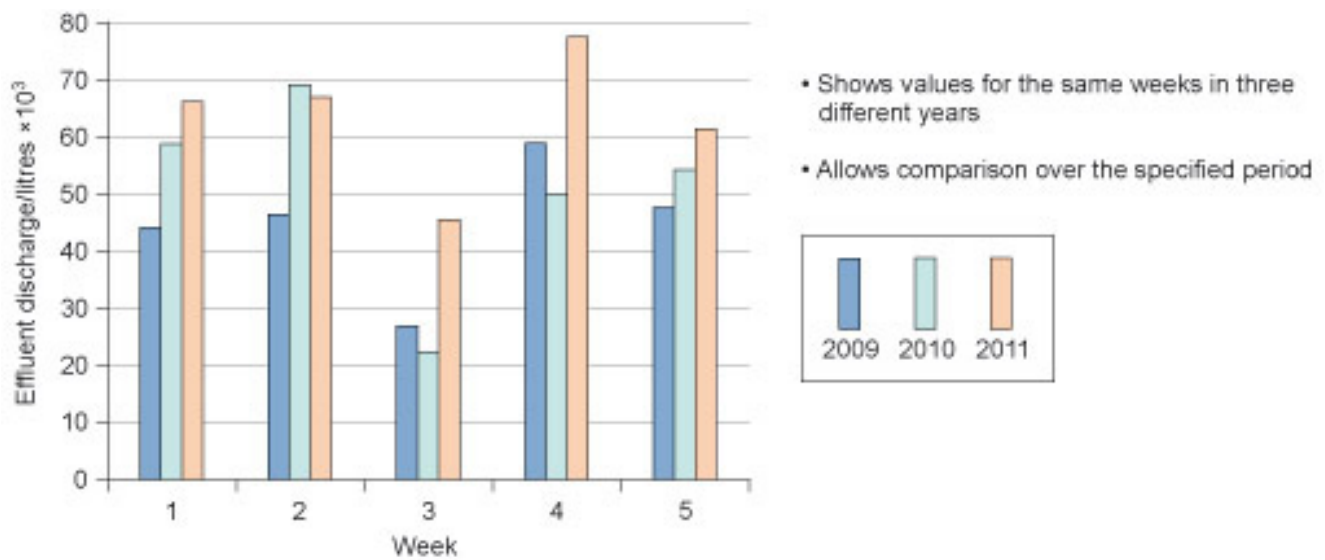


Figure 8 Example of a clustered bar chart: effluent discharges compared for weeks 1–5 over three years [Maximise](#)

Show description ▼

In Figure 8, it is clear what the relative values are over the three year periods. It is also reasonably easy to see what the three individual trends are, although less so than in a chart containing the values from only one year. Increasing the number of types of object being plotted in a single diagram will usually result in less clarity.

There are many other means of plotting the data: Figure 9 contains the same data as Figure 8, in a 3D plot. Although this plot might look impressive, in practice it is less informative than the simpler plot in Figure 8. For example, in Figure 8 it is straightforward to see the absolute values of the individual bars. This is much harder in Figure 9. Also, the relative heights of the bars are often less clear. In Figure 8 it is clear that the week 5 bars are increasing in size over the three years. This is much harder to see in Figure 9.

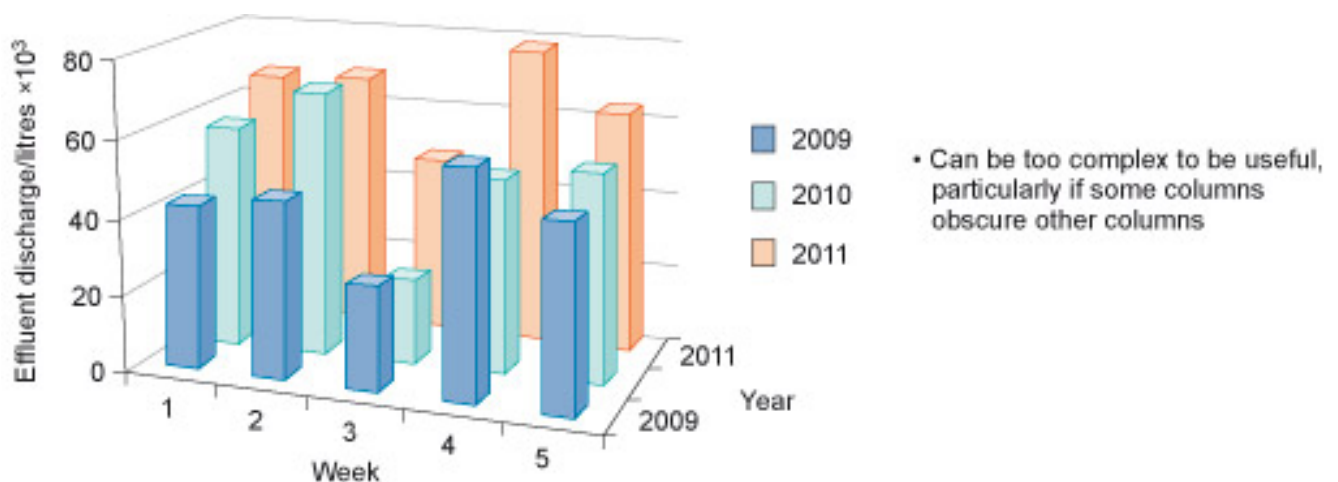


Figure 9 Example of a 3D clustered bar chart: effluent discharges compared for weeks 1–5 over three years [Maximise](#)

Show description ▼

Bar charts may also be presented horizontally (Figure 10), with the independent and dependent variables swapping axes. This format is often used in order to compare values of the dependent variable for several organisations or other entities in a particular month or year, rather than values for the same organisation over a number of months or years.

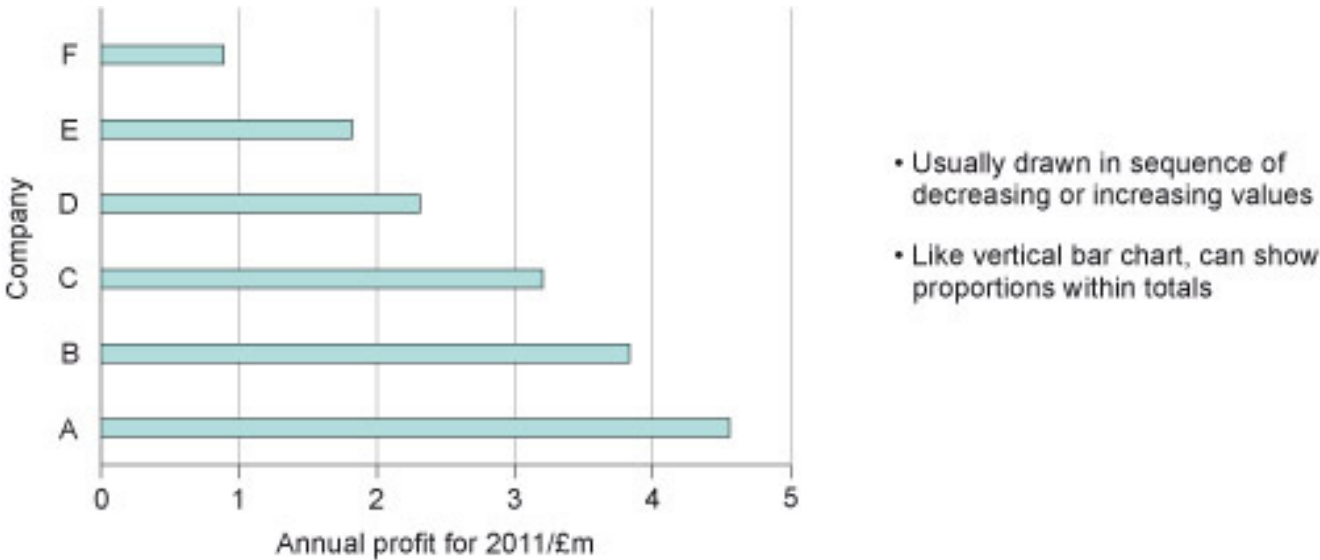


Figure 10 Example of a horizontal bar chart: annual profit for companies in 2011

[Maximise](#)

Show description ▼

Histograms

The main purpose of a histogram (Figure 11) is to show how samples of continuous data fall into each of the intervals under consideration. The most common and least common intervals can be seen clearly from the chart, as can the general shape of the sample distribution.

Height/cm	Width of interval, w	Number of plants, n	Frequency, n/w
0–4	4	9	2.25
4–6	2	19	9.5
6–7	1	27	27
7–8	1	24	24
8–12	4	13	3.25
12–20	8	8	1

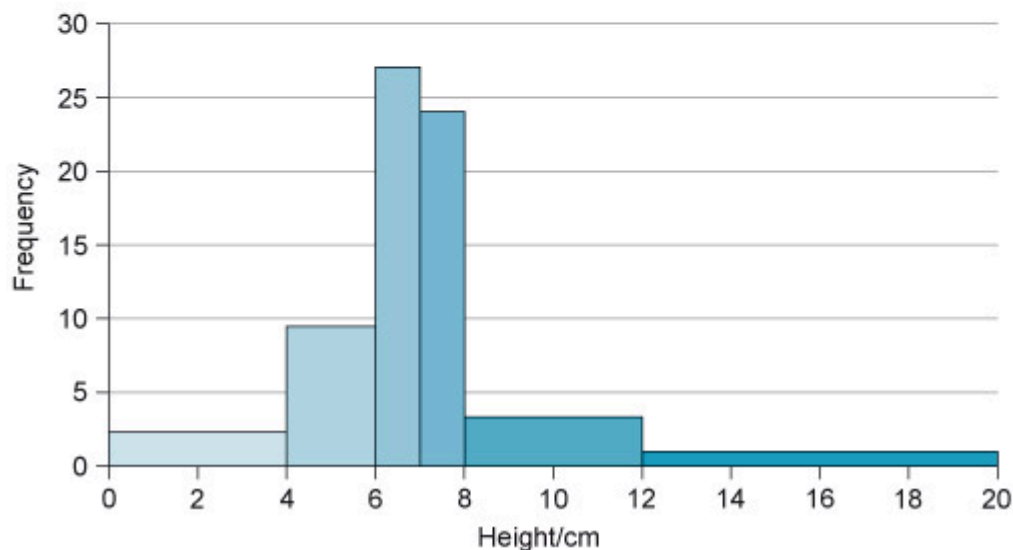


Figure 11 Example of a histogram: heights of 100 plants using fertiliser A

Show description ▼

Remember that it is area and not height that is important in a histogram. Thus if the areas of all the ‘bars’ are added, the total should be equal to the total number of samples taken. This may involve some calculation in order to draw the histogram correctly, as shown in the table in Figure 11.

Note that although the figure shows (for example) that a total of eight plants had a height of between 12 and 20 cm, it does not show where in this range they fell – it would be incorrect to assume that the 12–20 cm interval may be divided into smaller intervals of equal frequency. Thus choice of interval size in a histogram may have a significant effect on the information that may be gained from it.

3.3 Pie Charts

A pie chart (Figure 12) can be useful to show the proportions of different categories that make up a whole. However, pie charts are only really effective where there are few variables being compared, and they do not convey the absolute size of the segments. For this reason, it is uncommon to see pie charts being used in technical literature, where bar charts are more commonly used. A bar chart showing the data from Figure 12 is given in Figure 13.

Depending on the purpose of the chart it is usually the case that a bar chart is more informative.

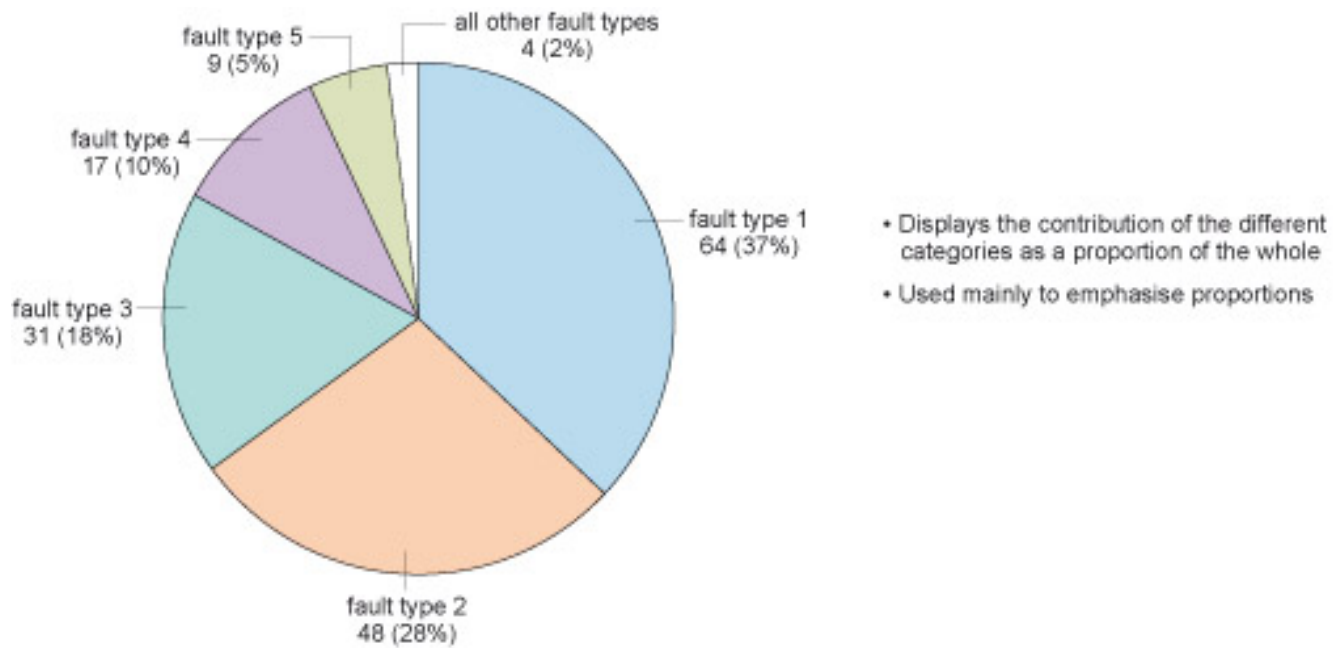


Figure 12 Example of a pie chart: proportions of different fault types

[Maximise](#)

[Show description](#)

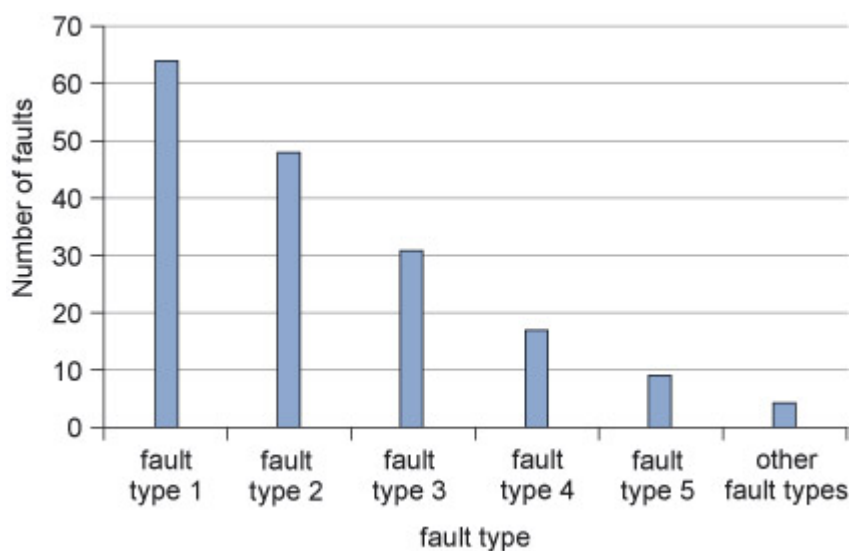


Figure 13 Example of a bar chart: absolute number of different fault types

[Show description](#)

3.4 Radar charts

A spider or radar chart (Figure 14) is useful for comparing two or more organisations, or one organisation on several different occasions. By plotting two charts on the same set of axes you can also compare, for instance, the budget for different departments against their actual spending. Figure 14 shows, from Year 1 to Year 2, a decrease in defective products and an increase in customer satisfaction and turnover.

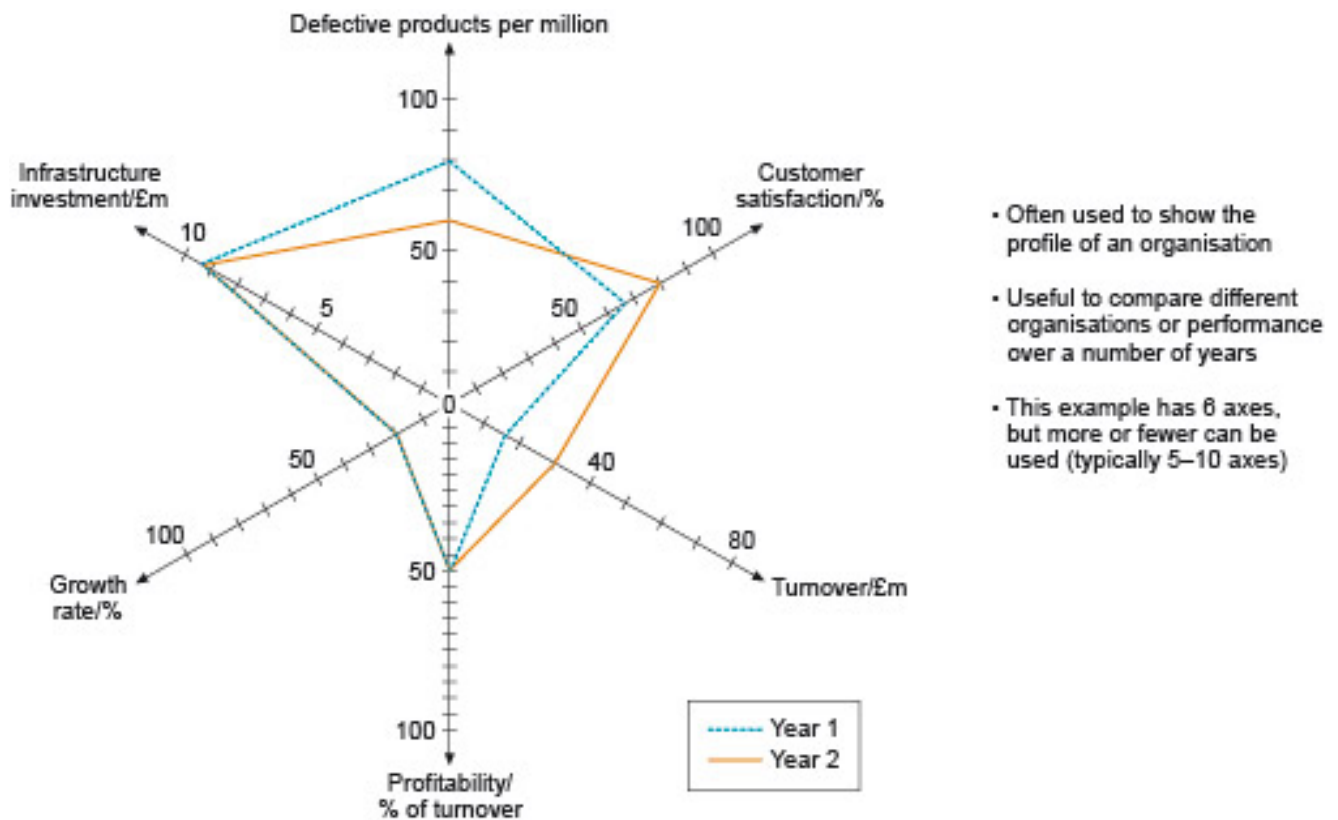


Figure 14 Example of a radar chart: profile of company performance for Years 1 and 2

[Maximise](#)

[Show description](#)

Activity 3

Go to Activity 3 on graphical representation of quantitative data.

4 Qualitative data acquisition

4.1 Analysing qualitative materials

Quantitative data implies the collecting of numerical data, yet technological research does not always deal solely with numbers and statistical calculations. Useful raw data might include:

- graphical data such as photographs, diagrams and sketches
- observations of how people work and react to situations
- observation of machines and mechanical processes
- observation of natural processes in the wider environment
- personal comments, both oral and written.

Any of these can potentially provide information and insights that help to achieve the objectives. Indeed, they form the typical components of case-study research. However, of particular importance is the

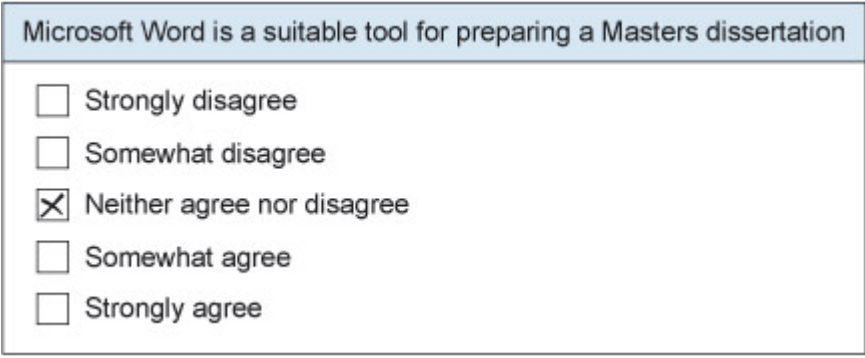
qualitative data provided by personal comments, which may range from casual chats to formal interviews and written reports.

I will not go into depth about how to analyse qualitative data from interviews, observations and general questionnaires here. These are potentially very difficult areas, and you are strongly advised to consult a suitable source for your methodology such as Silverman (2004) or Blaxter et al. (2006). When analysing qualitative data, you should bear in mind that a weak methodology, or general comments on unsystematic or arbitrary data are unlikely to provide the evidence required for a successful T802 project.

As an illustration, I will briefly discuss Likert tests, which many T802 students find useful for questionnaire-based research.

4.2 Likert tests

Many students working on case studies find it useful to ask respondents to use a **Likert scale** when canvassing opinions. When using a Likert scale, a respondent is given a list of statements, and asked to give the degree to which they agree or disagree with each one. Each statement in the list is called a **Likert item**; an example is given in Figure 15.



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☐ Strongly disagree

☐ Somewhat disagree

☒ Neither agree nor disagree

☐ Somewhat agree

☐ Strongly agree

Figure 15 Example of a 5-point Likert item for a survey question

Show description ▼

The response format for each Likert item is typically a 5-point or 7-point scale, where the middle point represents 'no opinion'. By using a well-designed set of Likert items, the respondents' opinions on an issue can be gauged.

Although Likert tests seem straightforward to use, there are several pitfalls that T802 students commonly fall into (a good, technical overview of these issues is given by Carifio and Perla (2007)).

1. Using too few items

A well-designed survey should use a range of Likert items to obtain a rounded view of the respondent's opinion. The 5-point scale typically used for a Likert item is a very coarse-grained tool for an individual question, particularly when the response might also be swayed by misinterpretation of the question, or different cultural or technological presumptions on the part of the respondent.

A similar danger is attempting to interpret a single response outside the context of the rest of the survey (although with a well-designed survey, this should not be a problem).

2. Using an inappropriate response format

A common mistake made by T802 students is to omit the middle point from the response list. For example, the survey might use a 4-point response list with options, ‘Strongly disagree’, ‘Somewhat disagree’, ‘Somewhat agree’ and ‘Strongly agree’. This is known as a ‘forced choice’ questionnaire. Survey writers sometimes feel that this is a useful technique to stop respondents from sitting on the fence. However, in T802 research, this attitude is quite inappropriate: your respondents may genuinely hold no opinion and should have the scope in the survey to express this. If you think that neutral responses will not give you the information that you require, you should reconsider the items in your survey, rather than force your respondents to answer in ways that they might consider incorrectly reflect their views.

Even worse is the use of an unbalanced response list; that is, one in which there are not an equal number of points either side of neutrality. Using an unbalanced response list runs the risk of skewing the data.

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☐ Somewhat disagree

☐ Slightly disagree

☒ Slightly agree

☐ Somewhat agree

☐ Strongly agree

Figure 16 Example of a forced choice, unbalanced Likert item – do not do this!

Show description ▼

The example in Figure 16 illustrates both of these problems. The respondent has not been allowed to state that (s)he has no opinion (which would presumably fall somewhere between ‘slightly disagree’ and ‘slightly agree’), and there are two options below this mid-point, and three options above it.

3. Interpreting responses as numerical values

When analysing the results of a Likert item from a survey, it is common to find students (and others!) attempting to scale the items to try to find a mean or a standard deviation. For example, you might be tempted to assign values to the possible-responses for the 5-point Likert item in Figure 15 as:

Response	Score
Strongly disagree	-2
Somewhat disagree	-1
Neither agree nor disagree	0

Somewhat agree	1
Strongly agree	2

Then if, say, three respondents selected 'Somewhat disagree' while six responded 'Strongly agree' you might conclude that the average response was:

$$[(3 \times -1) + (6 \times 2)] / 9 = 1$$

and so conclude that the average (perhaps representing the dominant view of the population?) is 1, or 'Somewhat agree'.

This attempted analysis makes no sense. What, for example, is a mean score of 1 supposed to mean (what are the units of opinion)? How is a response of 'Strongly agree' supposed to represent twice as much of what 'Somewhat agree' represents? Twice as much of what? Why not three, four or five times? Some respondents are more likely than others to respond at the extremes of the available responses; how does this sort of numerical approach compensate for that?

Bear in mind that although the data you collect have an inherent order, ('Somewhat disagree' and 'Strongly agree' both represent more agreement than 'Strongly disagree'), this does not mean that the responses can be placed at equally spaced points along some numerical scale. So do not assume, for example, that the difference between 'Somewhat disagree' and 'Neither agree nor disagree' is somehow the same as the difference between 'Somewhat agree' and 'Strongly agree'.

So when analysing responses to Likert tests:

1. Do not attempt to take a mean from some numerical coding of the responses. The mode is a usable statistic, if using the mode will give you a useful insight into the responses.
2. For the same reason, do not attempt to calculate a standard deviation. As the responses to Likert items are not, in fact, numerical, the standard deviation from a numerical encoding will not be meaningful. It is reasonable to talk about the range of responses (so you might write something like 'Responses ranged from 'Strongly disagree' to 'Somewhat agree', with most (55% of) respondents selecting 'Neither agree nor disagree').
3. The best way to present the responses is in a bar chart, with a separate bar for each item in the response set. Do not merge separate response items into single bars, and do not present the results as a histogram, or any other format which suggests that the results exist on a continuum.

Activity 4

Go to Activity 4 on the use of qualitative data.

5 Deriving conclusions from data: context and reliability

The issue of how certain you can be about your conclusions, and the extent to which you can generalise from them, is always dependent on the sample size you are able to work with in your data gathering. The size and scope of your sample will always be a compromise. Ideally, you should select a large number of samples possessing an appropriate range of features that represent the whole population being investigated. In practice, you may be severely limited by the constraints of time and money, particularly in T802 for which less than six months is allocated for data collection. Where you would ideally like to carry out ten case studies, you may be limited to three, with the consequent difficulty of having to select the three that can be classed as most representative of the target population. Consequently you need to be modest about your conclusions and whether they can be abstracted or generalised for a much larger population.

Some more general overviews of how you can use and interpret data to support your research can be found in general guides such as Moore (1999) or Potter (2006). A good work which focuses heavily on how to handle evidence to validate hypotheses is given by Sapsford and Jupp (2006).

Also, it is rarely (if ever) appropriate to conclude that your conclusions are absolutely 'proved' or 'disproved'. In almost all cases, the most that you can conclude is that your evidence achieves your research objectives to a greater or lesser degree. Indeed, it is quite possible that some evidence will support it and other evidence will contradict it. It is your job to determine how far your data goes in addressing your research aim, and explain that to your readers.

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